

Homework 4

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6-3:

$$V = \frac{\sin 3\phi}{\rho}, \quad \epsilon = 2.8 \epsilon_0$$

(a) Find E and P at point $A(1, 20^\circ, 4)$

$$\vec{E} = -\nabla \cdot V$$

$$\text{In cylindrical: } \nabla \cdot V = \frac{1}{\rho} \frac{\partial (\rho V_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial V_\phi}{\partial \phi} + \frac{\partial V_z}{\partial z}$$

$$\vec{E} = -\frac{\partial}{\partial \rho} \left(\frac{\sin 3\phi}{\rho} \right) \vec{a}_\rho - \frac{1}{\rho} \frac{\partial}{\partial \phi} \left(\frac{\sin 3\phi}{\rho} \right) \vec{a}_\phi - \frac{\partial}{\partial z} \left(\frac{\sin 3\phi}{\rho} \right) \vec{a}_z$$

$$\boxed{\vec{E} = \frac{\sin 3\phi}{\rho^2} \vec{a}_\rho - \frac{3 \cos 3\phi}{\rho^2} \vec{a}_\phi}$$

At $A(1, 20^\circ, 4)$:

$$\vec{E} = \frac{\sin 3(20^\circ)}{(1)^2} \vec{a}_\rho - \frac{3 \cos 3(20^\circ)}{(1)^2} \vec{a}_\phi$$

$$\boxed{\vec{E} = \frac{\sqrt{3}}{2} \vec{a}_\rho - \frac{3}{2} \vec{a}_\phi} \quad \text{V/m.}$$

$$\therefore \epsilon = \epsilon_r \epsilon_0$$

$$\text{since } \epsilon = 2.8 \epsilon_0, \epsilon_0 = 10^{-9}/36\pi \text{ (F/m)},$$

$$\epsilon_r = 2.8$$

$$\therefore \text{since } \vec{P} = \epsilon_0 \chi_e \vec{E} \text{ and } \vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

$$\vec{D} = \epsilon_0 \vec{E} + \epsilon_0 \chi_e \vec{E}$$

$$\vec{D} = \epsilon_0 \vec{E} (1 + \chi_e) \rightarrow 1 + \chi_e = \epsilon_r$$

$$\vec{D} = \epsilon_0 \epsilon_r \vec{E} \rightarrow \epsilon = \epsilon_0 \epsilon_r$$

$$\vec{D} = \epsilon \vec{E}$$

$$\vec{P} = \epsilon_0 \chi_e \vec{E}$$

$$= \epsilon_0 (\epsilon_r - 1) \vec{E}$$

$$= \frac{10^{-9}}{36\pi} (1.8) \left[\frac{\sqrt{3}}{2} a_y - \frac{3}{2} a_x \right]$$

$$\boxed{\vec{P} = 1.38 \times 10^{-11} a_y - 2.39 \times 10^{-11} a_x \text{ C/m}^2}$$

(b) find volume charge density (ρ_v) at A. and also compute (ρ_v) by using $\rho_v = -\nabla \cdot \vec{P}$.

$\uparrow$
 volumetric charge
 density due to
 polarization.

$$\rho_v = \nabla \cdot \vec{D}$$

$$\vec{D} = \epsilon \vec{E}$$

$$\vec{D} = (2.8) \left(\frac{10^{-9}}{36\pi} \right) \left[\frac{\sqrt{3}}{2} a_\rho - \frac{3}{2} a_\phi \right]$$

$$\vec{D} = 2.144 \times 10^{-11} a_\rho - 3.714 \times 10^{-11} a_\phi$$

~~$$\rho_v = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\frac{2.144}{\rho^2} \right) + \frac{1}{\rho} \frac{\partial}{\partial \phi} \left(\frac{-3.714}{\rho^2} \right)$$~~

$$\rho_v = \epsilon \nabla \cdot \vec{E} = \epsilon \left[\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\frac{\rho \sin 3\phi}{\rho^2} \right) + \frac{1}{\rho} \frac{\partial}{\partial \phi} \left(\frac{-3 \cos 3\phi}{\rho^2} \right) \right]$$

$$\rho_v = -\epsilon \frac{\sin 3\phi}{\rho^2} + \epsilon 9 \frac{\sin 3\phi}{\rho^3}$$

At $(1, 20^\circ, 4)$:

$$\rho_v = (2.8) \left(\frac{10^{-9}}{36\pi} \right) \left[\frac{-\sin 3(20)}{(1)^2} + 9 \frac{\sin 3(20)}{(1)^3} \right]$$

$$\rho_v = \left(\frac{2.8 \times 10^{-9}}{36\pi} \right) (4\sqrt{3}) = \boxed{1.715 \times 10^{-10} \text{ C/m}^3}$$

$$\vec{P} = \left(\frac{1.8 \times 10^{-9}}{36\pi} \right) \left[\frac{\sin 3\phi}{\rho^2} a_\rho - 3 \frac{\cos 3\phi}{\rho^2} a_\phi \right]$$

$$\begin{aligned} \rightarrow \rho_{pv} &= -\nabla \cdot \vec{P} \\ \rho_{pv} &= -\nabla \cdot \epsilon_0 \chi_e \vec{E} \end{aligned}$$

$$\rho_{fv} = -\epsilon_0 \nabla \cdot \vec{E}$$

$$= \left(\frac{-1.8 \times 10^{-9}}{36\pi} \right) \left[\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\frac{\rho \sin 3\phi}{\rho^2} \right) + \frac{1}{\rho} \frac{\partial}{\partial \phi} \left(\frac{-3 \cos 3\phi}{\rho^2} \right) \right]$$

$$= \left(\frac{-1.8 \times 10^{-9}}{36\pi} \right) \left[\frac{-\sin 3\phi}{\rho^2} + \frac{9 \sin 3\phi}{\rho^3} \right]$$

$$\boxed{\rho_{fv} = \left(\frac{-1.8 \times 10^{-9}}{36\pi} \right) (4\sqrt{3}) = -1.03 \times 10^{-10} \text{ C/m}^3}$$

6.4: Conducting sheets are located at $y=1$ and $y=3$ planes.
 $V(y=1)=0$ and $V(y=3)=50\text{V}$, find $V(y=2)$.
 Assume $\rho_v = \frac{y}{4\pi} \text{ nC/m}^3$ and $\epsilon = 4\epsilon_0$.

Poisson's Equation: $\nabla^2 V = \frac{-\rho_v}{\epsilon_0 \epsilon_r}$

$$\rightarrow \epsilon_0 \epsilon_r = \epsilon_0 \cdot 4$$

$$\rightarrow \rho_v = \frac{y}{4\pi}$$

$$\nabla^2 V = \frac{-y/4\pi \times 10^{-9}}{4\epsilon_0} = \frac{-y \times 10^{-9}}{16\pi \epsilon_0}$$

$$\frac{\partial^2 V}{\partial y^2} = \frac{-y \times 10^{-9}}{16\pi \epsilon_0} = \frac{-y \times 10^{-9}}{4 \cdot 16\pi \times 10^{-9}} = -\frac{y}{4}$$

$$\Rightarrow \frac{dV}{dy} = -\frac{9}{4} y^2 + A$$

$$\Rightarrow \frac{dV}{dy} = -9y^2 + A$$

~~$$\Rightarrow \frac{dV}{dy} = -9y^2 + A$$~~

$$\Rightarrow V(y) = -\frac{9}{8} \frac{y^3}{3} + Ay + B$$

$$\Rightarrow V(y) = -\frac{9y^3}{24} + Ay + B$$

at $V(y=1) = 0$:

$$0 = -\frac{9}{24} + A + B$$

$$A + B = \frac{9}{24} \quad \text{---(i)}$$

at $V(y=3) = 50$:

$$50 = -\frac{9(3)^3}{24} + A(3) + B$$

$$50 = -\frac{81}{8} + 3A + B$$

$$A + B = 0.375 \quad \text{---(i)}$$

$$3A + B = 60.125 \quad \text{---(ii)}$$

From calculator:

$$A = 29.875, B = -29.5$$

$$\star V(y) = -0.375y^3 + 29.875y - 29.5$$

At $y=2$:

$$V(y=2) = -0.375(2)^3 + 29.875(2) - 29.5$$

$$V(y=2) = 27.25 \text{ V}$$

6.10 In cylindrical co-ords $V(\rho=4) = 0$ and $V(\rho=12) = V_0$
 $\downarrow$ 4mm $\downarrow$ 12mm.

If $\vec{E} = -6 \text{ a}_\rho \text{ kV/m}$ at $\rho=8\text{mm}$, determine V_0 .

Laplace's Equation: $\nabla^2 V = 0$

$\therefore$ In cylindrical:

$$\nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) = 0 \Rightarrow \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) = 0$$

~~if $\rho \frac{\partial V}{\partial \rho} = A \Rightarrow \text{const}$~~
 ~~$\frac{\partial}{\partial \rho} (A) = 0$~~

$$\rho \frac{\partial V}{\partial \rho} = A$$

$$\frac{\partial V}{\partial \rho} = \frac{A}{\rho}$$

$$V(\rho) = A \ln(\rho) + B$$

At $V(\rho=4\text{mm}) = 0$

$$0 = A \ln(4\text{mm}) + B$$

$$B = -A \ln(4\text{mm})$$

At $V(\rho=12\text{mm}) = V_0$:

$$V_0 = A \ln(12\text{mm}) + B$$

$$V_0 = A \ln(12\text{mm}) - A \ln(4\text{mm})$$

$$V_0 = A \ln \left(\frac{12}{4} \right)$$

$$* A = \frac{V_0}{\ln(3)}$$

$$V_0 = A \ln(3)$$

Now; $\vec{E} = -\nabla V$

$$\vec{E} = -\frac{dV}{dr} \cdot \hat{a}_r$$

since; $\frac{\partial V}{\partial r} = \frac{A}{r} = \frac{V_0 / \ln(3)}{r} = \frac{V_0}{r \ln(3)}$

$$\vec{E} = -\frac{V_0}{r \ln(3)} \hat{a}_r$$

At $r = 8\text{mm}$, $\vec{E} = -6 \cdot \hat{a}_r \text{ kV/m}$:

$$6 \cdot 10^3 \cdot \frac{1}{r} = \frac{V_0}{8 \times 10^{-3} \ln(3)}$$

$$V_0 = (6 \times 10^3) (8 \times 10^{-3}) \ln(3)$$

$$V_0 = 48 \ln(3) = 53.73$$

6.13 Potential field is $V = (c_1 r^2 + c_2 r^{-2}) \sin 2\phi$

(a) show that V satisfies Laplace's equation.

$$V = c_1 r^2 \sin 2\phi + c_2 r^{-2} \sin 2\phi$$

$$\nabla^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\nabla^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} (c_1 r^2 \sin 2\phi + c_2 r^{-2} \sin 2\phi) \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2} (c_1 r^2 \sin 2\phi + c_2 r^{-2} \sin 2\phi)$$

$$= \frac{1}{r} \frac{\partial}{\partial r} \left(r \left(2c_1 r \sin 2\phi - \frac{2c_2 \sin 2\phi}{r^3} \right) \right) + \frac{1}{r^2} \frac{\partial}{\partial \phi} \left(c_1 r^2 2 \cos 2\phi + c_2 r^{-2} 2 \cos 2\phi \right)$$

$$= \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(2\rho^4 c_1 \sin 2\phi - 2c_2 \sin 2\phi \right) + \frac{1}{\rho^2} \frac{\partial}{\partial \phi} \left(\rho^2 c_1 2 \cos 2\phi + \rho^{-2} c_2 2 \cos 2\phi \right).$$

$$= \frac{1}{\rho} \left[\frac{4c_1 \sin 2\phi \rho + 4c_2 \sin 2\phi}{\rho^3} \right] + \frac{1}{\rho^2} \left[-4\rho^2 c_1 \sin 2\phi - 4\rho^{-2} c_2 \sin 2\phi \right]$$

~~$$= \frac{4c_1 \sin 2\phi}{\rho^2} - \frac{4c_2 \sin 2\phi}{\rho^4} - 4c_1 \sin 2\phi - 4c_2 \sin 2\phi$$~~

$$= 4c_1 \sin 2\phi + 4\rho^{-4} c_2 \sin 2\phi - 4c_1 \sin 2\phi - 4\rho^{-4} c_2 \sin 2\phi$$

$$= 0$$

$\therefore$ since $\nabla^2 V = 0$, it satisfies Laplace's equation

(b) Find c_1 and c_2 if $V = 50$ V and $|E| = 100$ V/m at $(1, 45^\circ, 1)$.

$$V = (c_1 \rho^2 + c_2 \rho^{-2}) \sin 2\phi$$

$$50 = (c_1 (1)^2 + c_2 (1)^{-2}) \sin 2(45^\circ)$$

$$50 = c_1 + c_2 \quad \text{--- (i)}$$

$$-\vec{E} = \frac{\partial}{\partial \rho} [(c_1 \rho^2 + c_2 \rho^{-2}) \sin 2\phi] + \frac{1}{\rho} \frac{\partial}{\partial \phi} [(c_1 \rho^2 + c_2 \rho^{-2}) \sin 2\phi]$$

$$= \left[\frac{2c_1 \rho - 2c_2}{\rho^3} \right] \sin 2\phi \cdot \hat{a}_\rho + \left[\frac{c_1 \rho + c_2}{\rho^3} \right] 2 \cos 2\phi \cdot \hat{a}_\phi$$

At $(1, 45^\circ, 1)$:

$$-\vec{E} = (2c_1 - 2c_2) \cdot \hat{a}_\rho$$

$$|\vec{E}| = 2c_1 - 2c_2$$

$$100 = 2c_1 - 2c_2$$

$$50 = c_1 - c_2 \quad \text{--- (ii)}$$

$$c_1 + c_2 = 50$$

$$c_1 - c_2 = 50$$

$$2c_1 = 100$$

$$c_1 = 50, c_2 = 0$$

6.15: (a) show that $V = V_0(1 - a^2/\rho^2)\rho \sin \phi$ (where V_0 is constant) satisfies Laplace's eq.

$$V = V_0 \rho \sin \phi \left(1 - \frac{a^2}{\rho^2} \right)$$

$$\nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2}$$

$$= \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial}{\partial \rho} \left(V_0 \sin \phi \left(\rho - \frac{a^2}{\rho} \right) \right) \right) + \frac{1}{\rho^2} \frac{\partial^2}{\partial \phi^2} \left(V_0 \rho \left(1 - \frac{a^2}{\rho^2} \right) \sin \phi \right)$$

$$= \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \left(V_0 \sin \phi \left(1 + \frac{a^2}{\rho^2} \right) \right) \right) + \frac{1}{\rho^2} \frac{\partial}{\partial \phi} \left(V_0 \rho \left(1 - \frac{a^2}{\rho^2} \right) \cos \phi \right)$$

$$= \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(V_0 \sin \phi \left(\rho + \frac{a^2}{\rho} \right) \right)$$

$$= \frac{1}{\rho} \left[V_0 \sin \phi \left(1 - \frac{a^2}{\rho^2} \right) \right] + \frac{1}{\rho^2} \left[V_0 \rho \left(1 - \frac{a^2}{\rho^2} \right) (-\sin \phi) \right]$$

$$= \cancel{\frac{V_0 \sin \phi}{\rho} \left(1 - \frac{a^2}{\rho^2} \right)} + \cancel{-\frac{V_0 \sin \phi}{\rho} \left(1 - \frac{a^2}{\rho^2} \right)}$$

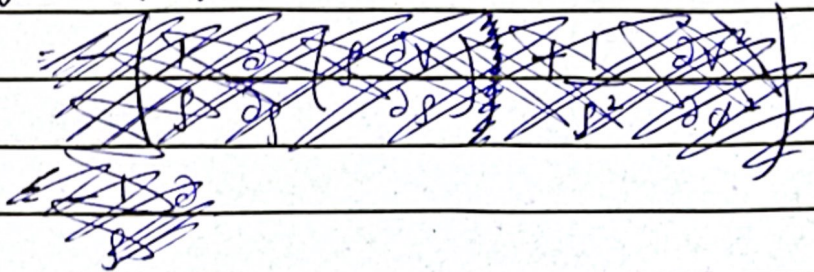
$$\boxed{\nabla^2 V = 0} \quad \text{Hence, Laplace is proved.}$$

(b) Determine $\vec{E}$ for $\rho^2 \gg a^2$.

If ρ^2 is much larger than a^2 then $\frac{a^2}{\rho^2}$ would approach 0.

$$V = V_0 \rho \sin \phi \left(1 - \frac{a^2}{\rho^2}\right) = V_0 \rho \sin \phi$$

$$\vec{E} = -\nabla \cdot V$$

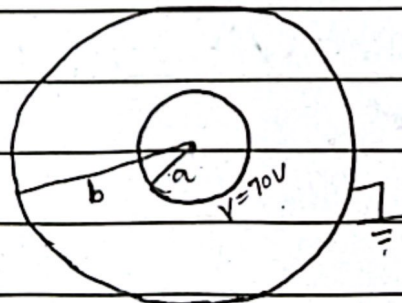


$$\vec{E} = -\frac{\partial V}{\partial \rho} \hat{\rho} - \frac{1}{\rho} \frac{\partial V}{\partial \phi} \hat{\phi}$$

$$= -\frac{\partial (V_0 \rho \sin \phi)}{\partial \rho} \hat{\rho} - \frac{1}{\rho} \frac{\partial (V_0 \rho \sin \phi)}{\partial \phi} \hat{\phi}$$

$$\boxed{\vec{E} = -V_0 \sin \phi \cdot \hat{\rho} - V_0 \cos \phi \cdot \hat{\phi}}$$

6.19:



$$a = 0.6 \text{ mm}$$

$$b = 30 \text{ mm}$$

(a) Find potential at $\rho = 15 \text{ mm}$

Since, all free charges move to surface inside a conductor $\nabla \cdot V = 0$ and Laplace's equation holds true.

$$\nabla^2 V = 0$$

$$\Rightarrow \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) = 0 \Rightarrow \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) = 0$$

$$\Rightarrow \rho \frac{\partial V}{\partial \rho} = A \Rightarrow \frac{\partial V}{\partial \rho} = \frac{A}{\rho} \Rightarrow V = A \ln(\rho) + B$$

$$\text{at } \rho = a, V = 70:$$

$$70 = A \ln(a) + B$$

$$\text{at } \rho = b, V = 0:$$

$$0 = A \ln(b) + B$$

$$B = -A \ln(b) \quad \text{--- (i)}$$

$$70 = A \ln(b) - A \ln(a)$$

$$70 = A \ln\left(\frac{b}{a}\right) \Rightarrow A = \frac{70}{\ln\left(\frac{b}{a}\right)} = \frac{70}{\ln(30/15)} = \frac{70}{\ln(2)} = 17.89$$

~~$$B = -A \ln(b) = -17.89 \ln(30 \times 10^{-3}) = 62.73$$~~

$$B = -A \ln(b)$$

$$= -17.89 \ln(30 \times 10^{-3})$$

$$B = 62.73$$

$$V = -A \ln(\rho) + B$$

$$V(\rho = 15 \text{ mm}) = -17.89 \ln(15 \times 10^{-3}) + 62.73$$

$$V(\rho = 15 \text{ mm}) = 12.4 \text{ V}$$

(b) If an electron is injected radially through a small hole in the inner electrode with velocity 10^7 m/s, find its velocity at $\rho = 15$ mm.

Potential Energy Gained = Kinetic Energy loss

$$Vq = \frac{1}{2} m (v^2 - u^2)$$

$$q_v (70 - 12.4) = \frac{1}{2} m ((10^7)^2 - v^2)$$

$$57.6 \times 2 \times q_v = 10^{14} - v^2$$

$$v^2 = 10^{14} - \left[\frac{57.6 \times 2 \times q_v}{m} \right]$$

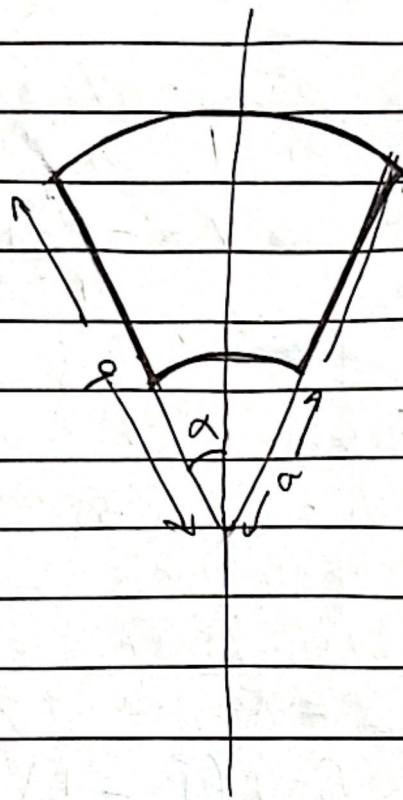
$$v^2 = 10^{14} - \left[\frac{115.2 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}} \right]$$

$$v^2 = 7.975 \times 10^{13}$$

$$v = \sqrt{7.975 \times 10^{13}}$$

$$v = 8.93 \times 10^6 \text{ m/s}$$

6.28: Show that the resistance of the sector of a spherical shell of conductivity σ , with cross section shown (where $\phi [0:2\pi]$), between its base from $r=a$ to $r=b$ is $R = \frac{1}{2\pi\sigma(1-\cos\alpha)} \left[\frac{1}{a} - \frac{1}{b} \right]$.



$$R = \frac{V}{I} = \frac{\int_L E \cdot dl}{\int_S \epsilon E \cdot ds}$$

From book method to find Resistance is given by:

1. Choose a co-ordinate system
2. Assume V_0 as the potential difference b/w conductor terminals.
3. Solve Laplace to obtain $V \rightarrow \vec{E} = -\nabla V \rightarrow I = \int_S \epsilon \vec{E} \cdot d\vec{s}$.
4. Obtain R as V_0/I .

* Assuming $V(r=a) = 0$ and $V(r=b) = V_0$.

$$\nabla^2 V = 0$$

In spherical co-ordinates:

$$\nabla \cdot V = \frac{\partial V}{\partial r} \cdot a_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \cdot a_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \cdot a_\phi$$

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$$

$$\Rightarrow \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) = 0 \Rightarrow \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) = 0$$

$$\Rightarrow r^2 \frac{\partial V}{\partial r} = A \Rightarrow \frac{\partial V}{\partial r} = \frac{A}{r^2} \Rightarrow V = -\frac{A}{r} + B$$

$$V(r=a) = 0$$

$$0 = -\frac{A}{a} + B$$

$$B = \frac{A}{a}$$

$$V(r=b) = V_0$$

$$V_0 = -\frac{A}{b} + B$$

$$V_0 = -\frac{A}{b} + \frac{A}{a}$$

$$V_0 = A \left(\frac{1}{a} - \frac{1}{b} \right)$$

$$A = \frac{V_0}{\frac{1}{a} - \frac{1}{b}}, \quad B = \frac{V_0}{\frac{1}{a} - \frac{1}{b}}$$

$$V = -\frac{V_0}{\frac{1}{a} - \frac{1}{b}} + \frac{V_0}{\frac{1}{a} - \frac{1}{b}} = \frac{V_0}{\frac{1}{a} - \frac{1}{b}} \left(\frac{a}{r} - 1 \right)$$

$$\vec{E} = -\nabla \cdot V$$

$$= -\frac{\partial V}{\partial r} \cdot \hat{a}_r = -\frac{\partial}{\partial r} \left(\frac{V_0}{\frac{1}{a} - \frac{1}{b}} \left(\frac{a}{r} - 1 \right) \right)$$

$$\vec{E} = \frac{V_0}{r^2 \left(\frac{1}{a} - \frac{1}{b} \right)} \cdot \hat{a}_r$$

$$\vec{J} = \sigma \vec{E} = \frac{V_0 \sigma}{r^2 \left(\frac{1}{a} - \frac{1}{b} \right)} \cdot \hat{a}_r$$

$$I = \int_C \vec{J} \cdot d\vec{S}$$

$$I = \int_{\theta=0}^{\alpha} \int_{\phi=0}^{2\pi} \frac{V_0 \sigma}{r^2 \left(\frac{1}{a} - \frac{1}{b} \right)} \cdot r^2 \sin \theta \cdot d\theta \cdot d\phi$$

$$I = \frac{V_0 \sigma}{\frac{1}{a} - \frac{1}{b}} \int_{\theta=0}^{\alpha} \int_{\phi=0}^{2\pi} \sin \theta \cdot d\theta \cdot d\phi$$

$$I = \frac{V_0 \sigma}{\frac{1}{a} - \frac{1}{b}} \cdot (-\cos \theta) \Big|_0^{\alpha} \cdot \phi \Big|_0^{2\pi}$$

$$I = \frac{2\pi V_0 \sigma (1 - \cos \alpha)}{\frac{1}{a} - \frac{1}{b}}$$

Ans: $R = \frac{V_0}{I} = \frac{V_0}{\frac{2\pi V_0 \sigma (1 - \cos \alpha)}{\frac{1}{a} - \frac{1}{b}}} = \frac{1}{2\pi \sigma (1 - \cos \alpha)} \left[\frac{1}{a} - \frac{1}{b} \right]$

Q.29 A spherical shell has inner and outer radii a and b , respectively. Assume that the shell has a uniform conductivity (σ) and that its copper electrodes plated on the inner and outer surfaces. Show that;

$$R = \frac{1}{4\pi \sigma} \left[\frac{1}{a} - \frac{1}{b} \right]$$

Taking Assumptions from 6.28 which were $V(r=a)=0$ and $V(r=b)=V_0$, we got the resistance for a sector such that;

$$R = \frac{1}{2\pi\sigma(1-\cos\alpha)} \left[\frac{1}{a} - \frac{1}{b} \right]$$

where α was the angle of sector.

In a spherical shell $\alpha = \pi$.

$$\text{Hence; } R = \frac{1}{2\pi\sigma(1-\cos\pi)} \left[\frac{1}{a} - \frac{1}{b} \right]$$

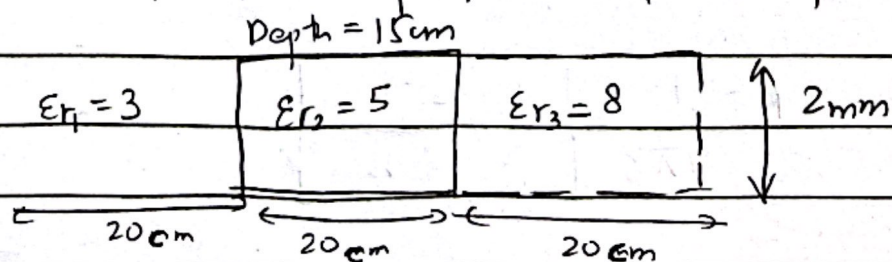
$$R = \frac{1}{2\pi\sigma(1-(-1))} \left[\frac{1}{a} - \frac{1}{b} \right]$$

Ans:

$$R = \frac{1}{4\pi\sigma} \left[\frac{1}{a} - \frac{1}{b} \right]$$

6.32

Calculate capacitance of parallel-plate capacitor shown



For a parallel-plate capacitor;

$$C = \frac{Q}{V} = \frac{\epsilon S}{d}$$

↳ S = Area of the plate

↳ d = distance between plates

↳ $\epsilon = \epsilon_0 \epsilon_r$

$$C = C_1 + C_2 + C_3$$

$$C = \frac{\epsilon_0 \epsilon_{r1} S}{d} + \frac{\epsilon_0 \epsilon_{r2} S}{d} + \frac{\epsilon_0 \epsilon_{r3} S}{d}$$

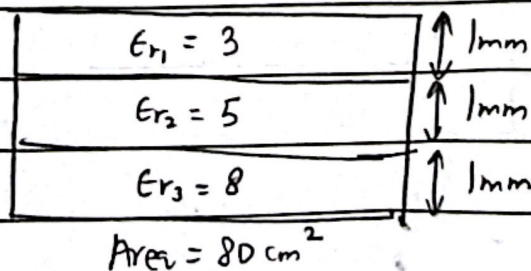
$$C = \frac{\epsilon_0 S}{d} [\epsilon_{r1} + \epsilon_{r2} + \epsilon_{r3}]$$

$$= \frac{10^{-9} (15 \times 10^{-2} \times 20 \times 10^{-2})}{36\pi (2 \times 10^{-3})} [3 + 5 + 8]$$

$$= \frac{10^{-9} (30 \times 10^{-3})}{36\pi (2 \times 10^{-3})} [16] 4$$

$$C = 2.122 \times 10^{-9} \text{ F} \approx 2.12 \text{ nF}$$

6.33: Evaluate capacitance of parallel-plate capacitor with multilayer dielectric shown.



Capacitor in series hence;

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$= \frac{d}{\epsilon_0 \epsilon_{r1} S} + \frac{d}{\epsilon_0 \epsilon_{r2} S} + \frac{d}{\epsilon_0 \epsilon_{r3} S}$$

$$= \frac{d}{\epsilon_0 S} \left[\frac{1}{\epsilon_{r1}} + \frac{1}{\epsilon_{r2}} + \frac{1}{\epsilon_{r3}} \right]$$

$$= \frac{1 \times 10^{-3} (36\pi)}{(1 \times 10^{-9}) (80 \times 10^{-4})} \left[\frac{1}{3} + \frac{1}{5} + \frac{1}{8} \right]$$

$$\frac{1}{C} = 1.414 \times 10^{10} (0.6583)$$

$$\frac{1}{C} = 9.306 \times 10^{-9}$$

$$C = 1.074 \times 10^{-10} \approx 0.1074 \text{ nF}$$